



Government R&D support's effects on export performance via innovation: An analysis of organizational motivators as moderators

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ABSTRACT

How and under what conditions does government R&D support lead to innovation and in turn increased export activities? We bridge these two relationships which are normally explored separately by creating a linking evolutionary economics based framework which uses variation and selection. We examine the relationship between government R&D support and export intensity, mediated by innovation, in a balanced panel of 289 innovative exporters in China. Leveraging the knowledge-based view, we also explore the roles of financial bonus incentives (an extrinsic motivator) and employee suggestions (an intrinsic motivator) as moderators to contribute to the debate on what type of motivators work best. This study contributes to the export and innovation literatures by showing that government support for R&D facilitates innovation and that innovation in turn facilitates exports. Further, the study shows that both extrinsic and intrinsic motivators moderate the staged relationships, but they affect different stages with bonuses moderating the R&D support – innovation relationship and suggestions moderating the innovation – export relationship.

1. Introduction

Many governments invest considerable resources in fostering firm innovation (Carvalho, 2018) and internationalization (Lederman, Olarreaga, & Payton, 2010) because these factors have been shown to be important determinants of firm performance (Love & Roper, 2015). However, innovation is a complex process (Crossan & Apaydin, 2010; Van de Ven, Polley, Garud, & Venkataraman, 1999), where higher input does not always lead to higher output (Saunila & Ukko, 2012). In other words, higher investment in R&D does not always result in higher innovation performance (Hall & Bagchi-Sen, 2002; Raymond & St-Pierre, 2010). Therefore, managing the innovation process rather than just ensuring enough input is vital to ensure high innovation performance and to gain a better understanding of the innovation process (Wolff, 2007; Rosenbusch, Brinckmann, & Bausch, 2011). The relationship between innovation and exports is similarly complex (Monreal-Perez, Aragon-Sanchez, & Sanchez-Marin, 2012). Existing research (e.g., Cassiman & Golovko, 2011; Harris & Li, 2011) has shown that innovation and exports are related in various ways, including

through product innovation (Rodríguez & Rodríguez, 2005) and process innovation (Alvarez, 2004; Rodríguez & Rodríguez, 2005), and potentially even in the form of a dynamic virtuous cycle (Golovko & Valentini, 2011). Therefore, there is a need to better understand the R&D-innovation-export relationship.

Innovation is expensive, but once done, the marginal cost of using the technology in a product or service is usually quite small (Romer, 1990). Thus, firms are often keen to sell products and services based on their innovations as widely as possible, both domestically and internationally, to recoup innovation costs and maximize profits to the fullest extent. Furthermore, a paper by Pla-Barber and Alegre (2007) demonstrated that the innovation-exports relationship is not a spurious correlation resulting from economies of scale/ firm size. Innovative firms often venture into foreign markets either due to their economic ambitions or in search of niche customers. It has been suggested that an important avenue to explore in developing a more complete understanding is to gain a better understanding of the microfoundations of innovation (Stam & Nooteboom, 2008) and firm internationalization (Chittoor, Aulakh, & Ray, 2019). Barney and Felin (2013) discuss and

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interpret that in management studies, microfoundations commonly refer to questions of social aggregation, including behavioral strategies and origins of capabilities and competitive advantage. One of the under-researched areas is the ability of firms to fully engage employees in core processes through motivation schemes. Initial research has identified motivation as crucial for achieving key firm goals which are explored in this study, namely firm innovation (Gupta, 2020) and exporting (Wood, Logar, & Riley, 2015), but much more research is needed on the topics to fully understand the relationships. To help increase our knowledge on the above-mentioned issues, this paper examines whether and how government R&D support leads to innovation and in turn increased exports.

Exports are defined in the Collins (2003) as “goods or services sold to a foreign country or countries,” and a common way to measure exports or export performance in the international business (IB) literature is export intensity which is calculated as the ratio of exports to total sales (Leonidou, Katsikeas, & Samiee, 2002) and is what we will use in this study. We define innovation following the classic Oslo Manual [of innovation] (OECD, 2018: 68) which defines innovation as “a new or improved product or business process (or combination thereof) that differs significantly from the firm’s previous products or business processes and that has been introduced on the market or brought into use by the firm.” R&D support is primarily targeted to benefit R&D activities. However, when the support is provided directly to firms, it is natural that firms want to convert the R&D activities into sales revenue including foreign sales. Ryan and Deci (2000a) discussed motivation as a combination of energy, direction, persistence and equifinality and summarize its importance in the real world in the short phrase: “Motivation produces” (p. 69). The most common way to consider different types of motivators is to break them down into those providing intrinsic motivation which Ryan and Deci (2000b: 55) write “refers to doing something because it is inherently interesting...and extrinsic motivation which refers to doing something because it leads to a separable outcome.” Comparing extrinsic and intrinsic motivators makes an interesting contrast. Love, Roper, and Zhou (2016) suggested that employee engagement could play a significant role in both innovation and exporting and extrinsic and intrinsic motivation facilitate engagement. Many firms establish motivation-related organizational arrangements to address the activation and intention aspects of employees (Ryan & Deci, 2000a).

Leveraging theories of extrinsic and intrinsic motivation, we explore the moderating effects of the use of financial incentives (financial bonus incentive schemes, an extrinsic motivator) and employee suggestions (an intrinsic motivator) in both stages of the mediated relationship (R&D support-innovation and innovation-exporting). This allows us to distinguish the specific factors facilitating innovation and export performance. By exploring the roles of intrinsic and extrinsic motivation, this paper also provides additional evidence to help untangle their relative roles in motivating employees, something prior research had considered unclear (Kuvaas, Buch, Weibel, Dysvik, & Nerstad, 2017). Stieglitz and Heine (2007) make a valuable contribution to our understanding of innovation by applying an evolutionary economics perspective. In this view, the corporate resource base generates innovation through constant ‘variation’, while the marketplace tests new products through ‘selection’. Both innovation and exporting, as well as government allocation of R&D support, are future-oriented activities where decisions are impacted by the track dependence stemming from the past. Therefore, integrating concepts from evolutionary economics into the innovation and exporting literature could provide a more comprehensive view of interconnected actions and mechanisms.

To advance the literature on exporting and innovation, we analyze the 289 most innovative exporting firms in China. China is the largest emerging market in terms of GDP and the second largest economy globally, directing significant government support for R&D to drive innovation and to become a global economic powerhouse. Indeed, innovation has become a top priority for China in recent years, and

China’s support for innovation has attracted much global attention. Therefore, it is important to explore the impact of government support on innovation and exports. Chinese firms and government initiatives supporting innovation have undergone significant changes in recent decades. Studies have observed China’s transition (Buckley et al., 2007; Child & Rodrigues, 2005) from labor-intensive, less sophisticated products and inward FDI policies to more R&D and outward market seeking strategies. Despite years of some engagement in R&D, international expansion remains challenging for Chinese firms due to lack of experience and/or weak reputation (Fey, Nayak, Wu, & Zhou, 2016), leading to the R&D/technology/intellectual property focus in China’s recent five-year plans. Chinese employees are demanding higher salaries which can only be justified through increased value addition, likely only achievable through innovation. By focusing on the performance of the most innovative firms, we aim to eliminate the effects of poor-quality manufacturing and lack of value proposition to foreign customers. The tension between selling in a very large domestic market versus seeking higher profitability internationally raises the research question: does government support for innovation lead to increased innovation output and a larger share of sales in foreign markets?

This research makes three significant contributions to theory and practice. First, few studies have explored how innovation supports firm exports and the role of government support for R&D in this process. Our study aims to bridge this gap by integrating the relationships between R&D support, innovation and exports. We extend and leverage a theoretical framework of two layers of variation-selection processes, inspired by Stieglitz and Heine (2007), to explain how R&D impacts export performance through the mediation by innovation. This unique perspective helps reshape our understanding of why export success rates for innovation-oriented firms may be low, utilizing the evolutionary economics concepts of variation and selection in a multi-stage model. Secondly, previous studies have not examined the roles of intrinsic and extrinsic motivations in facilitating innovation and export performance. Leveraging the knowledge-based view (KBV), we enhance our understanding of boundary conditions by showing that both intrinsic and extrinsic motivators moderate the mediated relationship between government R&D support and export performance. We find that each moderator only affects a specific section of the mediated relationship, shedding light on the specific boundary conditions. Extrinsic motivators enhance the government R&D support-innovation relationship while intrinsic motivators boost the innovation-export relationship. Thirdly, we offer new empirical evidence and practical implications for the government R&D support-export relationship. We find that R&D support increases export activities (via increased innovation) by an indirect-only mediation relationship. This presents evidence that government support mechanisms do not need to be targeted to specific areas to deliver socio-economic benefits, emphasizing the importance of studying regular and incidental spillovers to understand the broader range of indirect effects of government allocation of R&D support. This deepens our understanding of government R&D support policies that need a unified framework to account for their combined impacts on innovation and exports, two key drivers of economic growth.

2. Theory and hypotheses

2.1. A variation-selection view to government R&D support and exports

The nexus of innovation, technology institutions and economic dynamics along with their underlying interactions have been explored through theories of evolutionary economics (Malerba & McKelvey, 2019). The ideas of evolutionary economics are considered important, but there is no coherence in the literature as to what evolutionary economics refers to (Witt (2008). Generic concepts from evolutionary thinking have been applied to many areas, such as technology, language and economics (Witt, 2008). Alchian (1950) described similarities between biologists and economists in that they both seek to predict the

environmental chances of survival of living organisms, in which the participants act in their best interests but are not aware of the details of the global environment. In an approach, inspired by theories from biology and genetics attempting to describe common elements of all evolving systems (Hodgson, 2003), the key terms relevant to economics and strategy are variation, selection, and continuity or inheritance of information (Nelson & Winter, 1982; Stieglitz & Heine, 2007; Winter, 2014). In this terminology, variation refers to a purposeful increase in the number of options to increase the number of successful combinations and selection refers to a result of the reduction of surviving entities due to environmental factors.

The analogies between nature and economics are not perfect (Hodgson, 2003). Vromen (2004) argued that man-made evolution substantially differs from the processes in nature due to human creativity and social learning. This framing allows the discussion of economic processes with their implications over time, especially if the phenomena rely on creating increased variation (such as the allocation of government support or exploratory innovation activities). Alchian (1950) discussed that there are at least two sources of inter-firm variation: uncertainty due to differing judgment and opinion, and trial-and-error from intentional as well as unintentional invention. Knudsen (2002) called this type of variation 'migration' and noted that whereas entry and exit are often assumed, migration is largely ignored in the literature. Yet, this is what leads to innovation and new products being tested in the markets. Innovation plays a role in a firm's overall strategic choices. For example, Kaul (2012) argued that technological innovations cause the subsequent reorganization of a firm's portfolio in order to redeploy scarce resources to new areas of opportunity. Possibilities that were previously out of reach become feasible after successful innovations. As the earlier rent-maximizing equilibrium is altered, new markets and niches, including export markets, become possible and promising after innovation (Kaul, 2012). The inheritance analogy is based on assets built (e.g., in the example of plants by Nelson & Winter, 1982) or loosely defined habits or routines developed over time (Hodgson, 2003).

2.2. The innovation-export cycle

Insights gained modelling national economies emphasize the importance of technological progress ahead of capital deepening (Solow, 1957). However, since technological innovation inherently creates disequilibria in the models, recent developments have attempted to address these disequilibria by applying an evolutionary perspective (Iwai, 2000). Multiple authors have provided strong evidence that innovation is linked to exports. Studies suggest that the relationship between innovation and exports is unidirectional, either from innovation to exports (Cassiman & Golovko, 2011) or from exports to innovation (Bustos, 2011), while others propose it is bidirectional (Golovko & Valentini, 2011; Monreal-Perez et al., 2012). An intriguing question arises regarding whether there is an inherent conflict between exploration/innovation and exploitation/export (Prashantham & Birkinshaw, 2012) or if these phenomena are complementary (Harris & Li, 2011; Hennart, 2020). Despite this, there is limited research on the mechanisms that interrelate them or the origins and facilitating factors of the innovation-export cycle.

Harris and Li (2011) observe that while macroeconomics literature has established several solid theoretical frameworks for trade and innovation relationships (neo-endowment models based on specialization on labor, capital, technology; and neo-technology models that link innovative industries to net exporting), microeconomic research on innovation as a determinant in internationalization is primarily empirical and lacks a strong foundation. Some commonly referenced papers, like Cassiman and Golovko (2011) or Golovko and Valentini (2011) only examine the binary states (yes/no) of exporting and innovation or focus solely on the initial export decision (e.g., Harris & Li, 2011), neglecting export growth over time. Bustos (2011) introduced a three-stage model

of firm evolution that sequences the milestones of the innovation-export cycle, with profits increasing at each stage: first, firms run low technology operations and do not export; second, firms remain involved in low technology operations but start to export; third, firms become involved in high technology operations and continue to be an exporter. Kafourous, Buckley, Sharp, and Wang (2008) support this view, arguing that firms need to be active in multiple markets to maximize returns from their innovations. However, in a sample of the most innovative firms, such as the ones we studied, the process of export and innovation may occur in reverse order, with firms becoming more innovative before starting to export. This reverse sequence was also noted by Cassiman and Golovko (2011) who linked higher level of exporting to increased firm productivity. The innovation-to-export approach may be the most viable growth strategy for firms in small open economies (Kyläheiko, Jantunen, Puumalainen, Saarenketo, & Tuppur, 2011) and emerging economies with underdeveloped industrial sectors (Singh, 2009). Innovation impacts export performance on various levels. We focus on the most commonly used export performance measure in the international marketing literature which is export intensity (Leonidou et al., 2002).

2.3. Export performance and government R&D support: a framework of variation and selection procedures

Exporting has not received much attention in the literature in recent decades compared to other more complex and high-commitment entry modes like international joint ventures or wholly-owned subsidiaries. However, exporting is important to understand since it is the most common way firms initially start doing business in another country because compared to higher commitment entry modes like international joint ventures or wholly owned subsidiaries exporting involves lower levels of commitment, risk, and complexity (Lu & Beamish, 2006). As a result, exports are the preferred initial way of internationalizing (Young, Hamill, Wheeler, & Davies, 1989).

The literature on the determinants of export performance points to a long list of variables that play a role: from firm characteristics, management characteristics, and market characteristics to strategies that are employed (Chen et al., 2016; Sousa, Martínez-López, & Coelho, 2008; Zou & Stan, 1998). Instead of developing a general model, empirical studies tend to cover a few fragmented aspects which explain only part of export success. Chen, Sousa, and He's (2016, p. 627) meta-analysis refers to several papers which support their view that "a systematic theoretical basis and framework that could comprehensively explain all of the drivers of export performance remains absent". Export performance studies do not commonly treat government support as a separate factor but list as one of the external factors. Nevertheless, it is recognized that government support can increase the capacity of firms to overcome export challenges and take up more risky internationalization investments that firms would otherwise not undertake (Torres, Clegg, & Varum, 2016). Chetty and Hamilton's (1993) meta-analysis of firm-level determinants of export growth categorized three types of drivers: firm characteristics, firm competencies, and export strategy. The strongest individual determinant was 'technology', listed under firm competencies. Indeed, Yang, Chen, and Chuang (2004) show that a firm's level of technology positively affects exporting. The above supports our approach to investigate the extent to which innovation activities lead to foreign sales as well as the knowledge focus on internationalization via exports. Knowledge has a two-way relationship with export activities. For example, Geldres-Weiss, Uribe-Borquez, Coudounaris, and Monreal-Perez (2016) argue that the route to export performance begins by first learning-by-exporting and accumulating knowledge to build a diverse product or service portfolio which in turn results in firms leveraging the knowledge gained to open up a variety of new export markets.

Stieglitz and Heine (2007) suggest that more radical innovations may require new resources. The government, with its socio-economic interests aligned with firm success, can be a provider that supports such

development. The individual relationships between government R&D support and innovation, and innovation and exports, are well established in the R&D policy and management literatures. Government support for export activities, especially from the perspective of SMEs, is a fairly well-studied connection (Alvarez, 2004). However, there is a lack of research on their combined effect, in other words, between government R&D support and exports. Using Irish data Görg, Henry, and Strobl (2008) found that grants can help firms that are already exporting, but there is no evidence that they encourage non-exporting firms to start exporting. Guo, Guo, and Jiang (2016) studied the Chinese Innofund R&D program and found that it increased exports by 2.4–2.6 % and sales of new products by 7.9–11.7 %, depending on the grant size. Although the positive effect on exports is noteworthy, Wang, Li, and Furman (2017) found that receiving an Innofund grant had no effect on firm survival, patenting, or venture funding. Furthermore, Hong, Feng, Wu, and Wang (2016) suggested that the efficiency of Chinese government R&D grants depended on the industry, while the efficiency of private R&D funds had a positive impact on patent applications in all sectors.

We address the research gap by examining the probabilistic relationship between government R&D support and exports at the firm level. We explain this relationship using the theoretical concepts of evolutionary economics within a framework of two distinct layers of variation and selection processes, a novel approach not found in prior literature. In the language of evolutionary economics, ‘variation’ refers to an increase in the number of scenarios leading to a desired outcome, while ‘selection’ refers to a narrowing down of possible scenarios due to social or economic processes (Knudsen, 2002). Hence, when a government selects recipients for R&D support, it introduces more ‘variation’ without causing immediate ‘selection’.

When a government allocates R&D support to firms, it pursues a double goal: to support innovation, and to generate increased revenue and socio-economic benefits from the products of the innovation. Creating opportunities for increased variation does not mean that the resource allocation is nondirected or random (Alchian, 1950). However, governments face uncertainty about which firms will best deliver desired results. Therefore, in the initial phase, the government introduces inter-firm variation to maximize the likelihood of achieving end goals. Subsequently, firms that receive government support generate additional intra-firm variation. This variation arises from the different strategic choices made by the firm, the specific knowledge employed within the firm for innovation development and the mix of incentives used to motivate employees. As a result, the breadth and quality of the resulting patent portfolio will vary, leading to differing

probabilities of success among firms.

The function of creating additional variation serves to increase the overall odds for favorable outcomes for some firms in the model. At the same time, the selection process becomes more evident. Some variation support beneficiary firms and lay the foundations for new product development and exports in firms’ R&D activities, while other variations render firms unable to deliver any innovation outcome (the first round of the selection). This does not mean these firms will exit from the market, but they become less likely to export their new products. Finally, innovation and exports can be expected to be correlated. For example, Baldwin and Harrigan (2007) showed that the higher the product quality, the more distant markets a product can be sold to. However, even for successful patentees/innovators, there will be a second round of selection by the market as foreign markets present a significant hurdle even for innovative products. A schematic representation of our conceptual model is presented in Fig. 1.

2.4. Mediation of innovativeness

Tidd’s (2001) review of decades of innovation management literature concluded that despite ample research, it has failed to produce an ideal ‘best practice’ model. According to the author, it is difficult to establish firm level empirical links between innovation inputs and outputs, and especially between innovation and performance. The research on innovation has been based on a wide range of disciplines, providing many insights but lacking a common framework (Tidd, 2001). Furthermore, Tidd (2001) argues that the lack thereof is caused by various environmental contingencies and organizational configurations. These factors remain largely hidden by agencies allocating R&D support but are likely to determine export success to a great extent.

As government support is provided to specific firms, it is important to explain how government support can impact export performance at the firm level to establish causal relationships. In complex phenomena like this, multiple causal links can be organized in stepwise models to describe the interplay and timing. While there are a few stepwise innovation models for productivity, we have not found similar models that directly include innovation inputs and exporting activities. However, Cassiman and Golovko (2011) have demonstrated that innovation is linked to exports. Therefore, we suggest that the innovation mechanisms at work may follow similar to the three steps outlined by Crepon, Duguet, and Mairesse (1998): (i) the decision-making on innovation inputs, (ii) the knowledge production function and (iii) the relationship between innovation output and productivity; and four steps by Halpern and Murakozy (2012): (i) decision on R&D investment, (ii) decision on

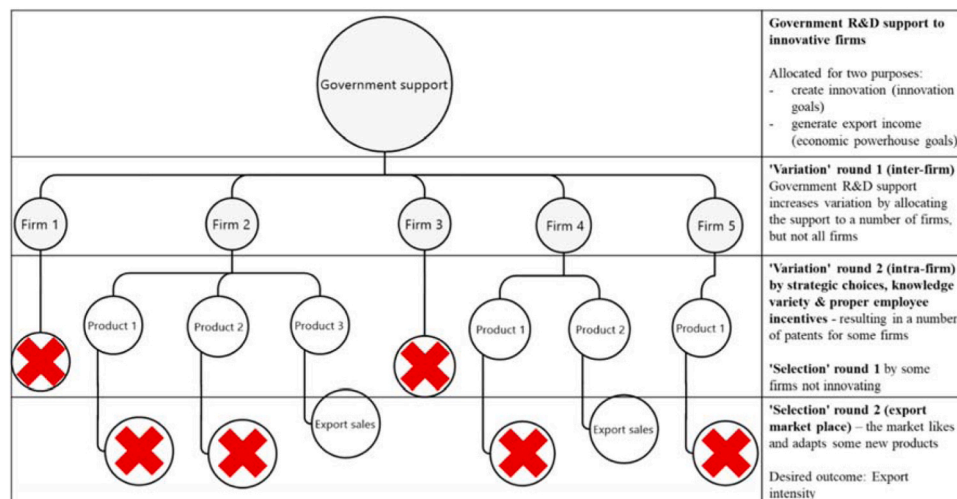


Fig. 1. Theoretical model of ‘variation’ (increasing the number of scenarios towards the desired outcome) and ‘selection’ (decreasing the number of scenarios towards the desired outcome) rounds after allocating government R&D support.

the level of R&D, (iii) R&D transformation to innovation output and (iv) innovation transformation to productivity.

Research has shown that government support for R&D leads to positive innovation output. In our analogy, we relate decisions on inputs to the application and acquisition of government R&D support, which is then transformed to innovation output. Following the principles of evolutionary economics, we use an innovation output measure as our mediating variable to capture an increased variety (rather than the quality) of potentially marketable products and services. In the second step, we draw on literature suggesting that more productive firms can more easily overcome export barriers (e.g., Bernard & Jensen, 1995; Melitz, 2003). In short, we propose a pathway where government R&D support influences innovativeness, which then affects export intensity.

Empirical observations support our theoretical proposition. In countries with open trade policies, export markets are theoretically open to all firms. However, in practice, only a fraction of firms actually export. Low export intensity is also evident in our sample of the most innovative Chinese firms. Exporting poses a challenge for many emerging market firms even if they are engaged in innovation and product development. The two-layer variation-selection conceptual model that we have developed and presented above offers some explanations for the generally low success rate in foreign markets. Many governments provide direct targeted support to the most promising firms in order to increase the likelihood of achieving desirable outcomes such as innovation and export success. Even with a low probability of a significant breakthrough, small export earnings contribute to the domestic economy and generate socio-economic benefits.

We argue that more innovative firms can similarly overcome export barriers and achieve better export performance that allows for foreign markets as a selection scenario. Therefore, we hypothesize that there is a stepwise relationship between government support for R&D and export intensity through innovation. We hypothesize that:

Hypothesis 1. *Government support for R&D is positively related to export intensity (H1i) through the mediating effect of innovativeness (H1ii).*

2.5. Knowledge as a competitive advantage

The purpose of Hypothesis 1 is to establish the baseline mediated relationship, but simply receiving government R&D support does not automatically result in innovation or exports. Knudsen (2002, p. 458) argued that Nelson and Winter's (1982) discussion of the 'organizational gene' was a crucial contribution to the literature. What is missing is input from employees, or in other words, consideration of knowledge factors. Grant (1996) argued in a seminal article, which established the KBV, that knowledge resides in individuals and the main role of an organization is to facilitate knowledge application. He stated, "it is in the analysis of coordination within the firm that the KBV promises to make its biggest contribution" (Grant, 1996: 121). In our mediated model, knowledge application is most evident in innovativeness. The innovativeness node is also instrumental in resolving some knowledge boundary issues by combining individual knowledge and applying it for the benefit of the firm, as well as addressing whether the knowledge is public or private.

Markey-Towler (2018) developed a formal psychological model showing that innovation generation requires functioning rules and processes to turn ideas into knowledge, and promote the evolutionary growth of knowledge. Chen et al. (2016) noted that the export literature primarily explores the impact of knowledge through organizational learning theory. Hitt, Bierman, Shimizu, and Kochhar (2001) argue that the relationship between human capital and firm strategies is more complex than generally assumed, including factors like professional reputation, tacit knowledge and social capital. Knowledge needs to be governed effectively to benefit the firm. Nickerson and Zenger (2004) differentiate between three types of governance choices for supporting internal knowledge formation: markets, authority-based hierarchies,

and consensus-based hierarchies. Building on this distinction, they suggest that the correct alignment of problem complexity and governance method does not guarantee a solution, but the approach works statistically over a large number of solution searches. However, Blomqvist and Kianto (2015) point out that knowledge that is not applied in everyday practice, such as shelved patents or high competencies of employees, does not create value for the firm. Therefore, it is crucial to find a way to fully utilize the useful knowledge the firm possesses. Following Grant's (1996) premise that knowledge is primarily within employees, employee motivation is a vital component of a firm's knowledge strategy.

Theories of evolutionary economics and determinants of export performance are relevant for macro or meso level analysis. The description of heterogeneous agents in evolutionary economics has emphasized the importance of including human motives in discussions of innovation (Gerschlager, 2012). Motivation is arguably best understood at the individual level, with intrinsic motives like intellectual challenge and extrinsic motives like personal financial incentives playing a role in enhancing employee motivation without interaction with each other (Kominis & Emmanuel, 2007). Further, Amabile (1997) argues that managers need to be more aware of how to foster intrinsic motivation as it is essential for creativity, but laments that evidence mostly comes from controlled laboratory experiments, due to the challenges of its objective measurement.

The dichotomy between intrinsic and extrinsic motivation is well established in the organizational behavior (OB) literature and informs the basis of choosing the variables in our theoretical model. We explore how intrinsic and extrinsic rewards influence firm level outcomes in the stages of our mediated relationship. Employee motivation is an important focus for psychology and organizational behavior researchers with implications for individuals and sub-organizational units, despite being rarely discussed in the IB literature. Individual motivation can impact firm-level outcomes as trusting relationships encourage employees to invest in firm-specific human capital benefiting the firm (Wang, He, & Mahoney, 2009). Motivation programs in firms address aspects of motivation, such as energy, direction, persistence and equifinality (Ryan & Deci, 2000a) by indicating the desired direction and end goal, rewarding effort and persistence. According to Wang et al. (2009), understanding employee willingness could be the key to unlocking employee underinvestment. Nevertheless, there is no relevant literature about selecting the appropriate construct to measuring intrinsic and extrinsic motivation at the firm level (a similar observation was made by Kim, Pathak, and Werner (2015) who developed a custom survey for measuring collaborative culture). While extrinsic motivation can be easier to proxy at the firm level due to the formalized compensation packages, the schemes vary greatly between firms and there are no standardized accounting categories which separate additional pay from regular salary.

2.6. Moderation by employee participation

In business settings, innovation is not created externally by receiving R&D support but is turned into successful innovation (patenting) via internal processes. Grant (1996) notes that despite a general understanding of the importance of coordination within the firm, there is a lack of a rigorous integrated widely accepted theory of how this works. He posits that the essence of a team-based organization is the direct involvement of specialists, which is beyond the ability of managers to effectively coordinate. In terms of inputs, innovation is primarily a product of intellectual efforts, not one of machinery or capital.

Receiving government support for R&D does not ensure that it will result in innovation nor does having an innovative product or service ensure that it will be exported. Both present opportunities and increase the likelihood of success. Greater resources for R&D increase the chances that one can produce an innovative output, but hard work is required by employees to turn R&D resources into an innovative outcome.

Likewise, a more innovative product provides a higher chance/better opportunity that the product can be exported since it is more unique. Turning that potential opportunity into reality requires much hard work by a firm's employees. Highly motivated employees can help to achieve difficult tasks like this and both intrinsic (employee suggestion systems) and extrinsic motivators (bonuses) have been shown to help with employee motivation (Kominis & Emmanuel, 2007). However, intrinsic motivators such as employee suggestion systems are frequently suggested to be the most important motivator and have been shown to be especially important for motivating employees in innovative/creative tasks (Amabile, 1997) which both innovation and exporting are. We know from activation theory (Gardner & Cummings, 1988; Scott, 1966) that employees perform better when they have a higher level of activation which can be reached by providing them more interesting tasks that engage them more like making suggestions which employee suggestion systems facilitate. Furthermore, we know from Lindsley (Gardner & Cummings, 1988; Lindsley, 1951) that humans function better at this higher level of activation because the central nervous system, and especially the cerebral cortex, function better in such cases. Thus, introducing intrinsic motivators like employee suggestion systems can help increase the effectiveness (moderate) of processes like turning government support for R&D into innovation and turning innovation into successful exports. Similarly, research on employee engagement has shown that employee engagement (like employee suggestion systems) facilitates job performance (Dalal, Baysinger, Brummel, & LeBreton, 2012).

Ostensibly, the greater the variety of innovations the firm works on, the higher the chances of success. When a firm receives an R&D grant, the funding only offers a tool for the firm. Employees need to be motivated to act to turn the additional resources for R&D into an innovation. Innovation can be developed in a designated department or across business units and the innovation team can be small or large, but it is generally accepted that using ideas from non-managerial employees positively impacts firms' innovation performance (Andries & Czarnitzki, 2014). A source of ideas and facilitator of innovation is employee suggestions, creating intra-firm 'variation'. Buech, Michel, and Sonntag (2010, p. 507) assert that, "suggestion systems play a critical role in organizations because they save costs and channel innovative behavior in a useful direction." As a common weakness of suggestion systems is the difficulty in motivating employees to submit ideas (Fairbank & Williams, 2001), the mere opportunity to submit suggestions is usually an insufficient motivator. Therefore, instead of just proxying the existence of a system, we analyze its use by employees. Furthermore, at the stage of submitting a suggestion, it is difficult to assess its quality and contribution to the firm's sales, as this does not depend on the perceptions of experts or coworkers, and will be determined by the market. We argue that it is the process of involving the employees that on average improves the innovation performance of the firm per unit of government R&D funding, as the pool of potential ideas is very large and employees experience is critical to innovation performance. The higher the number of suggestions submitted, the more variety of product or service development is created, and the more potential the firm has to turn funding into tangible innovation. Therefore, we propose that a higher rate of employee involvement leads to improved innovation results from the R&D grant funding received:

Hypothesis 2a. *Higher rate of employees submitting suggestions positively moderates the relationship between government support to R&D and innovativeness.*

Traditionally innovation is associated with exploration as firms' efforts are directed towards new activities with unknown end results. Even the most innovative and highest quality new products and services may fail in markets, if timing, packaging, marketing strategy or for other external factors are not right. This is similar to the evolutionary view, where it is not always the strongest 'gene' that carries continuity as even the best combinations can be eliminated in the following generation due

to adverse events or environmental circumstances. Zang (2018) discusses exploitative innovation, which involves improving existing products and is closer to the existing market. Employees, who are often in closer contact with customers daily, may be more able to contribute to exploitative innovation, which can help firms expand their product reach to new markets. As shown in the previous section, fully engaged employees who make suggestions in the firm convert (positively moderate) innovation into exports in a more efficient system. This is due to a higher level of activation which benefits employee motivation and in turn employee performance driving progress in exporting.

Participation, such as making suggestions for how the firm can improve, appeals to employees' intrinsic motivation by offering intellectual challenges and fostering a sense of belonging. Von Krogh (1998) distinguished between high care and low care organizational relationships in knowledge creation, advocating for high care relationships where firms show they value employees. Implementing an employee suggestion system is one way to show care for employees, helping them develop trust and commitment to the organization, facilitating knowledge sharing and thus innovation and other desirable outputs such as exporting. In high trust and commitment relationships, innovation's effect on exporting is easier (positively moderates), as there are fewer process losses and more focus on facilitating exporting and not worrying if one can have trust in the firm. In low care relationships, employees hoard knowledge, rather than sharing it voluntarily (Von Krogh, 1998). Chiva and Alegre (2009) summarized a different categorization of employee engagement in their Organizational Learning Capability measurement scale, which incorporates five dimensions: experimentation, risk taking, interaction with the external environment, dialogue between employees and managers, and participative decision making. Fernández-Mesa and Alegre (2015) found that, in an Italian and Spanish sample, this construct was positively related to export intensity, showing that employee engagement such as using employee suggestion systems facilitates exporting.

Von Krogh (1998) discussed Polanyi and Prosch's (1975) notion of indwelling, a process of creating social knowledge that shifts individuals' perspectives from "looking at" to "looking with". This concept was extended to the business context by rephrasing it to: from "looking at the customer with his problems" to "looking with the customer at his problems" (p.141). Addressing the concerns of export customers are ostensibly more challenging than those of domestic clients, as if viewed through an evolutionary framework, products and services must pass a stricter selection process. Understanding the preferences of customers who are physically and culturally distant arguably requires a deeper understanding of their needs to maximize innovation exploitation. This necessitates a higher level of care within the organization to view problems 'with' customers and fully comprehend their perspective. Incorporating suggestions from diverse employees, especially those who interact with current and potential customers, enables the firm to better grasp customer preferences and effectively translate innovation into successful exporting. Gathering ideas from a wide range of employees, rather than just top managers, results in more original ideas that can enhance the process of transferring innovation into successful exporting, especially when targeting new potential foreign customers. The number of employee suggestions can be seen as a reflection of superior customer care and an indication of the increased variation of ideas to enhance products and services for end-users. Therefore, we hypothesize that:

Hypothesis 2b. *Higher rate of employees submitting suggestions positively moderates the relationship between innovativeness and export intensity.*

2.7. Moderation by financial incentives

Government support for R&D activities is commonly directed at different types of R&D related expenditures, such as purchasing laboratory equipment and hiring additional researchers. Positive impacts on innovation output are typically expected as a result. Government

agencies operating with limited budgets normally aim to generate the greatest returns for the support provided. Additionally, incentive systems within firms can lead to greater efficiency in innovation development either directly by motivating employees to work harder or longer or indirectly by attracting superior talent (Zenger & Lazzarini, 2004). Empirical research has shown that financial rewards have a positive effect on innovation (e.g., Chang, Fu, Low, & Zhang, 2015) and learning-by-exporting (Kang, 2020). We explore the effect of financial incentives, operationalized as the firm's preferred and used incentive being a bonus payment, since this is the most commonly used extrinsic incentive with the most straightforward short-term linkages to financial incentive theories. Establishing bonus systems that reward individual performance and align with overall firm goals can promote organizational integration through "collective bonds of performance" (Ghoshal & Gratton, 2002).

In the first stage of our mediated relationship, we are studying the connection between government R&D support and innovativeness. Amabile (1993) initially articulated that her experimental research demonstrates that extrinsic incentives can harm creativity, but they do not hinder work on technical quality. However, more recently Amabile (Hennessey & Amabile, 2010) has suggested that extrinsic rewards can actually have a positive effect on creativity, or in evolutionary economic terms, increasing the variation of ideas. Further, Camerer and Hogarth's (1999) review of experimental studies concludes that while financial incentives may reduce variance in risk-seeking situations, they improve performance in simple effort-responsive tasks, such as judgment, prediction, and memory recall. Innovativeness can be stimulated by both extrinsic and intrinsic incentives, but especially when measured by tangible outputs like patents as we do, a concerted long-term effort focusing on technical details is beneficial. Thus, we hypothesize that:

Hypothesis 3a. *Financial incentives positively moderate the relationship between government support for R&D and innovativeness.*

Finally, innovation does not guarantee success in the marketplace, especially in more competitive or challenging export markets. The arguments presented in the previous section relating to activation theory (Scott, 1966) and employee engagement more broadly also help explain why bonuses can positively moderate the relationship between innovation and exports. For a firm looking to sustain its operations, improving the efficiency of monetizing innovations is crucial, and financial motivation schemes aimed at employees play a role in achieving that goal. While it is reasonable to assume that financial incentives lead to increased sales by sales employees at the individual level, the correlation is not always evident at the firm level, as bonus payments for employees in various positions within the firm may be based on different performance metrics than direct sales. Wheatley and Doty (2010) found that bonus compensation moderates the relationship between innovation and firm performance, challenging the common belief that long-term equity type incentives are necessary for innovation strategies to succeed. While the Chinese context may differ, as its economy becomes more market-based, the effects of incentives will likely be similar to those effects observed elsewhere (Park, Li, & Tse, 2006).

The contribution of financial incentives to the innovation-export stage of the entire mediated relationship is not straightforward. Financial incentives tied to direct sales often do not differentiate between export sales and domestic sales. However, in a sample of innovative Chinese firms, foreign markets often offer better prices for their products and additional growth opportunities (Guan & Ma, 2003). Bonus oriented compensation incentives may motivate employees to pursue sales growth in an increased variety of markets, especially in riskier and more lucrative export destinations. Chinese firms in R&D-intensive industries are particularly interested in more demanding export markets, such as G7 countries (Feng, Li, & Swenson, 2016). Therefore, it is likely that in our innovation-oriented sample, bonus schemes implicitly or explicitly motivate sales to higher-margin export markets.

In conclusion, bonuses should help translate tangible innovation outputs into exports through motivated employees. To understand the export pathways of innovative firms, we aim to test a comprehensive empirical model and thus we hypothesize that:

Hypothesis 3b. *Financial incentives positively moderate the relationship between innovativeness and export intensity.*

3. Data and methods

3.1. Data and sample

We utilized data from two secondary datasets to test our hypothesis in the Chinese context. We selected China because the Chinese government is making significant efforts to enhance firm innovation in order to help Chinese firms become more competitive globally and increase their exports. The first dataset we use is the *Innovation-Oriented Firms Database* collected by China's Ministry of Science and Technology (MoST) from 2008 to 2011, which has a low non-response rate and can be accessed by a limited number of people. The MoST assesses if a firm meets the criteria to receive a subsidy, such as meeting the minimum threshold for R&D intensity, patents developed, or service innovations created in the previous three years. In the study, an 'innovative exporter' does not refer to a firm implementing innovative strategies for export markets but to a firm that develops new products primarily for domestic use and subsequently exports them. Firms that met the criteria are eligible for a government grant and are required to complete the survey.

The dataset contains detailed operational, financial, and innovation information from 400 leading innovating firms (across 14 two-digit industries in 31 provinces), serving as the primary source for all variables except export intensity. As far as we know, this dataset is one of the most detailed innovation surveys of the leading innovating firms in China. The majority of firms in the dataset are large enterprises, as 75 % of firms have over 1000 employees. Export data used to calculate export intensity was obtained from the Chinese Customs Trade Statistics database covering all exporting firms in China. The two datasets were merged for analysis. As we could not be sure that the non-exporters in the dataset do not use resellers or other intermediaries to export, we keep the firms that export for at least one year within the four years in our dataset yielding 289 firms and a balanced panel of 1156 firm-year observations, or four observations per firm.

To account for potential endogeneity, we lagged the independent variable for government R&D support. Acknowledging the self-selection bias in government support through the application-based approval process, we recognize that firms qualifying for assistance likely share similar profiles of merit and shortcomings (for example, Torres et al., 2016 found that firms seeking pro-internationalization government support often have financial constraints and seek to transfer risks to external partners). These are common challenges in any study of government support.

3.2. Measures

3.2.1. Dependent variable

In our study, we propose a parsimonious model with variables that are inherently defined by KBV principles and align with the logic of evolutionary mechanisms. In our context, international markets serve as a selection scenario to assess the competitiveness of knowledge-based products and services offered by a firm. Our dependent variable *export intensity* is an objective measure of export performance. It is calculated as the ratio of exports to total revenues, a common metric used in the IB literature (Leonidou et al., 2002). This metric not only reflects the competitiveness of the firm's products and services in international markets, but also indicates the level of the firm's level of knowledge about foreign markets (Ling-Yee, 2004). Export intensity has a low mean value (10.2 %), suggesting that, on average, 10 % of sales are

international. This indicates that export growth is challenging for the firms in our sample, and that the innovation in the most innovative firms in our sample may not necessarily lead to success in export markets due to selection forces.

3.2.2. Independent variable

While previous studies have often treated government R&D support as a binary event, we aim to investigate the effect of the degree of R&D support by measuring our independent variable *government_R&D* using the logarithm of the total amount of government R&D support. The R&D investment from the government reported in the survey includes investments from national labs, national key projects, and other government-supported projects. Government grants are allocated to support knowledge activities based on the firm's assessed potential for breakthroughs.

3.2.3. Mediating variable

We measure the mediating variable *innovativeness* using a commonly used innovation output measure: the number of patents per employee (Quintane, Casselman, Reiche, & Nylund, 2011). Patents not only indicate the innovativeness of the firm at a specific time (Ahuja & Katila, 2001) but also provide market protection from imitation and may enhance a company's negotiating position with partners (Blind, Edler, Frietsch, & Schmoch, 2006). Patents can also help mitigate liabilities of origin as Shu, Wan, Gao, and Liu (2014) discovered in a sample of Chinese firms where reputation was found to be the most important patenting motive.

Our patent data was collected through the MoST survey and reflects invention patents applied for in the surveyed year. In China's context, the invention patents represent higher quality patents compared to utility patents and other types. While there is an ongoing debate about whether patent count is a suitable measure for innovation in China, due to local government patent subsidy programs (Li, 2012), or if it remains relevant as it correlates with R&D input and financial output (Dang & Motohashi, 2015), but like many we use it as the best objective measure available and one of the most commonly used measures of innovation in existing literature.

3.2.4. Moderator variables

In our models, we utilize two moderator variables. The first variable is internally submitted suggestions per employee, which accounts for employee involvement (H2a and H2b). The second variable reflects the usage of bonus incentives (H3a and H3b). These variables refer to the contextual environment that impacts both the process outcome of innovativeness (H2a and H3a) and the performance outcome of export intensity (H2b and H3b). The variable *suggestions* is a continuous variable that expresses the degree of employee involvement, calculated from survey data on total suggestions divided by the number of employees in the company. Employee suggestions should enhance process efficiency and product and service quality. The variable *bonus_incentive* is a binary variable that indicates a preference for bonus incentive systems to reward employees for their innovation activities within the firm. It is marked as '1' if a bonus was the preferred motivation system used in the firm (ahead of promotion, equity incentives, or no incentive). This choice of variable is also based on several characteristics that consider behavioral biases. Firstly, risk and reward prospects are framed compared to some relative level. Incentives are only effective if they are not already factored into the base salary and are perceived as a fair bonus in comparison to coworkers and historical levels (Ockenfels, Sliwka, & Werner, 2015). Bonus systems offer the unassumed reward and incentives are accessible to a larger number of employees than the career promotion, which depends on the availability of senior positions and office politics. Secondly, loss aversion causes individuals to prefer sure payouts over risky prospects (Tversky & Kahneman, 1992). In most cases, people want immediate and frequent appreciation, rather than future payoffs. Therefore, bonus payments create the more immediate

link to the effort than equity incentives that provide later and more uncertain rewards. Thirdly, it is important to distinguish between various motivation systems as due to mental accounting, individuals tend to treat money from different sources separately (Thaler, 1999). Firm-level survey data has limitations but may be the most reliable source of information regarding the strategic use of bonus motivation within firms.

Conversely, the level of intrinsic motivation remains hidden at the firm level by most objective measurements. One method of estimating intrinsic motivation is by observing employee participation. In most cases this benefits from a qualitative evaluation, but in firms with formalized suggestion systems, these systems can provide a reasonable proxy for the intrinsic motivation through employee participation. While the scales of the moderator variables are not directly comparable, they reflect the degree of intrinsic and extrinsic motivation in the firm that influences the mediated relationship of the main model.

3.2.5. Control variables

In all models, we control for several factors that may influence a firm's patenting performance and export intensity: firm *size*, measured by the log of total assets; firm *age*, measured by the log of firm age; export experience, indicated by a dummy variable for firms with over eight years of exporting experience; *profitability* measured by return on assets to capture the quality of management activities; whether the firm is publicly *listed* as listing on a stock market requires more transparency and adherence to certain required or voluntary management principles; *R&D intensity* (the ratio of R&D expenditure to total revenue); and a binary indicator for whether the firm has used *foreign R&D*. Additionally, we control for the *year*, the *industry*, and *region* fixed effects.

An explanation of the variables and their descriptive statistics can be found in Table 1. Variance inflation factors (VIFs) are also reported in Table 1, all of which were below the recommended threshold value of 10, indicating that multicollinearity is not likely a problem. Our conceptual model is presented in Fig. 2.

4. Results

4.1. Regression results

In our analysis we conducted a series of OLS regressions. The results can be found in Table 2. First, we examined H1 in Models 1–3 in Table 2—focusing on the mediated relationship via innovation. We discovered that government R&D support does not directly affect export intensity significantly ($b = 0.001$, $p = 0.848$). When discussing mediation effects, it is crucial to note that theoretical literature suggests a mediated relationship can exist even without a confirmed direct relationship (Ager & De Boeck, 2017). Zhao, Lynch, and Chen (2010) label this scenario as indirect-only mediation in their typology of mediated relationships. In practice, this means that the relationship between the main independent variable and the main dependent variable depends on the context and is expected only if the independent variable leads to an increase in the mediating variable. We found evidence that government R&D support positively affects innovation (H1i) ($b = 0.139$, $p = 0.000$) and that innovation positively impacts export intensity (H1ii) ($b = 0.038$, $p = 0.041$). When evaluating results, it is important to consider not only the significance level but also the effect size (Fey, Hu, & Delios, 2023). A 1 % increase in the nominal value of government R&D subsidy received by the firm would lead to a 0.139 % increase in the number of patents applied for per 100 employees. Additionally, a 1 % increase in the number of patents applied for per 100 employees would result in a 3.8 % increase in export intensity. A Sobel test for the presence of mediation in H1i and H1ii resulted in $z = 1.929$, $p = 0.054$, confirming the presence of indirect-only mediation of innovation in the relationship between government R&D support and export intensity providing support for H1.

Next, we explored moderating relationships. In Model 4 in Table 2,

Table 1
Descriptive statistics and variable definitions.

Variable	Description of the variable	Obs	Mean	Std.Dev.	Min	Max	VIF
Export_intensity	the ratio of exports to total revenue	1156	0.102	0.169	0	1	
Government_R&D	ln(1 + total government R&D)	1156	0.715	0.802	0	5.583	1.61
Innovativeness	ln(1 + applied invented patents per 100 employee)	1156	0.424	0.390	0	1.261	1.21
Suggestions	ln(1 + the number of suggestions per employee)	1156	0.158	0.230	0	0.683	1.02
Bonus_incentive	'1' for using cash bonus as the first choice of incentive	1156	0.691	0.454	0	1	1.02
Size	ln(total assets)	1156	12.412	1.918	7.089	17.583	2.26
Age	Ln(firm age)	1156	2.749	0.596	0	4.357	2.00
Export_experience	'1' for firms with more than 8-year exporting experience	1156	0.616	0.487	0	1	2.00
ROA	return on assets	1156	0.075	0.081	-.513	0.503	1.04
Listed	'1' for listed firms	1156	0.543	0.498	0	1	1.24
R&D_intensity	the ratio of R&D expenditure to total revenue	1156	0.053	0.072	0	1.487	1.23
Foreign_R&D	'1' for having foreign R&D investment	1156	0.130	0.336	0	1	1.15

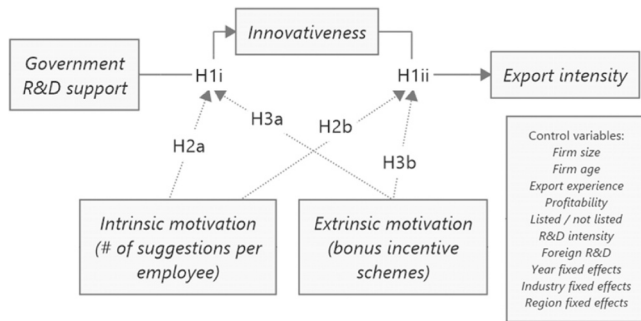


Fig. 2. Conceptual model. Note: moderating relationships are represented with a dotted line.

H2a which suggested that employee participation in suggestion systems moderates the relationship between government R&D support and innovation, was not supported ($b = -0.036$, $p = 0.551$). However, we did find support for H2b, which examined the effect of the same moderator on the relationship between innovativeness and export intensity ($b = 0.179$, $p = 0.015$). The moderating role of suggestion systems in the relationship between innovativeness and export intensity is depicted in Fig. 3a.

Lastly, we analyzed the moderation effects of financial incentives. For H3a, we found that the relationship between government R&D support and innovation is significantly moderated by bonus systems. This relationship is reasonably stable in both the purely H3a model ($b = 0.123$, $p = 0.048$) and the full model ($b = 0.141$, $p = 0.071$). However, for H3b, we found no evidence of moderation by bonus financial incentives in the relationship between innovativeness and export intensity ($b = 0.014$, $p = 0.696$). The moderating role of financial incentives in the relationship between government R&D support and innovativeness is depicted in Fig. 3b. This finding is somewhat counterintuitive as financial incentives could be expected to impact exploitation activities. Nevertheless, the support for H3a indicates that financial incentives are effective for the success of exploration activities.

The full model results are displayed in Models 7 and 9 for each step, demonstrating the consistency of our results, and in Fig. 4. Additionally, the control variables (presented in the upper section of Table 2, along with year and industry effects) remain the same and consistent across all models. In summary, the influence of government R&D support on export intensity occurs through mediation by innovation. The initial stage of the mediated model, from government R&D support to innovation is moderated by bonus financial incentives (extrinsic motivation). The subsequent stage of the model, from innovation to export intensity, is moderated by employee participation in making suggestions (intrinsic motivation).

4.2. Robustness tests

In line with Zhang, Wedel, and Pieters (2009), we utilize the Bayesian mediation approach to assess the robustness of our mediation analysis and ensure the robustness of our results. The bootstrap test results reveal that the indirect effect of the mediation is significant ($p = 0.009$), confirming the robustness of our findings. Second, to address endogeneity concerns, we implement a two-stage least square (2SLS) model for instrumental variables analysis. Following Beck, Lopes-Bento, and Schenker-Wicki (2016), we employed the average government R&D support by industry and region as the instrumental variable. An instrumental variable should be correlated with the IV, but not with the dependent variable. The rationale behind our selection of instrumental variable is that the average government R&D support is strongly correlated with firm-level government R&D support but not directly linked to firm-level innovation. The outcomes in Table 3 demonstrate that the mediation effects remain consistent with our results, with Sobel test for the presence of mediation in H1i and H1ii yielding $z = 1.858$, $p = 0.063$, indicating that our results are robust after controlling for endogeneity concerns. Finally, we test the robustness of our findings using alternative measures, employing the share of government R&D support to total R&D support as a proxy for alternative government R&D support and total granted patents per 100 employees as an alternative measure of innovativeness. The results presented in Table 4 are also in line with our initial findings.

5. Discussion and conclusion

Our study contributes to the export literature by adding rarely discussed perspectives that contribute to a holistic model of firm internationalization. We show that government support for R&D affects innovation which in turn affects exports. While the combination of innovation and export outcomes has been explored in previous literature, it has not been examined in a two-stage model or in a model exploring moderators as we do. The exploration of the partial relationship that does exist is primarily empirically focused and often lacks strong theoretical grounding. As other authors (e.g., Harris & Li, 2011) have indicated, models in macroeconomic contexts overlook the microeconomic point of view commonly used in international business. We build a stronger theoretical bridge between the exporting literature and the innovation literature, leveraging insights about microfoundations gleaned from evolutionary economics and economic psychology. Our study makes three key contributions.

First, few studies have explored how investment in R&D facilitates exports. We develop a theoretical framework leveraging two layers of variation-selection processes to explain the process of turning government support for R&D into export activities via innovation. This exercise, which extends evolutionary economics, may help develop an overarching theory of government support for internationalization. Governments, unable to predict well which firms will succeed, create variation by allocating support to a large number of companies. The

Table 2
Results of regression analysis.

Hypothesis	(1) H1	(2) H1	(3)	(4) H2a	(5) H3a	(6) Full model	(7) H2b	(8) H3b	(9) Full model
Dependent Variables	export-intensity	innovativeness	export-intensity	innovativeness	innovativeness	innovativeness	export-intensity	export-intensity	export-intensity
Size	− 0.010 (0.005) [0.024]	− 0.086 (0.009) [0.000]	− 0.007 (0.005) [0.137]	− 0.067 (0.011) [0.000]	− 0.066 (0.011) [0.000]	− 0.068 (0.011) [0.000]	− 0.011 (0.005) [0.024]	− 0.011 (0.005) [0.025]	− 0.011 (0.005) [0.025]
Age	0.004 (0.011) [0.729]	− 0.029 (0.027) [0.290]	0.005 (0.011) [0.649]	− 0.025 (0.025) [0.320]	− 0.020 (0.025) [0.434]	− 0.019 (0.025) [0.459]	0.009 (0.010) [0.373]	0.008 (0.010) [0.437]	0.009 (0.010) [0.379]
Export_experience	− 0.005 (0.018) [0.762]	0.039 (0.036) [0.279]	− 0.008 (0.018) [0.662]	0.036 (0.035) [0.313]	0.036 (0.035) [0.309]	0.037 (0.035) [0.295]	− 0.008 (0.017) [0.633]	− 0.011 (0.017) [0.536]	− 0.008 (0.017) [0.636]
ROA	0.131 (0.039) [0.001]	0.073 (0.067) [0.275]	0.128 (0.039) [0.001]	0.099 (0.066) [0.136]	0.097 (0.067) [0.148]	0.098 (0.068) [0.151]	0.144 (0.038) [0.000]	0.152 (0.039) [0.000]	0.144 (0.038) [0.000]
Listed	0.032 (0.013) [0.013]	0.030 (0.028) [0.279]	0.032 (0.013) [0.014]	0.009 (0.029) [0.753]	0.008 (0.028) [0.785]	0.007 (0.028) [0.791]	0.026 (0.013) [0.038]	0.026 (0.013) [0.044]	0.026 (0.013) [0.040]
R&D_sales	0.358 (0.106) [0.001]	0.569 (0.230) [0.013]	0.345 (0.105) [0.001]	0.652 (0.229) [0.004]	0.716 (0.241) [0.003]	0.738 (0.253) [0.004]	0.362 (0.103) [0.000]	0.360 (0.106) [0.001]	0.359 (0.103) [0.000]
Foreign_R&D	− 0.042 (0.013) [0.002]	0.113 (0.040) [0.005]	− 0.046 (0.013) [0.000]	0.093 (0.039) [0.018]	0.097 (0.039) [0.013]	0.098 (0.039) [0.013]	− 0.041 (0.013) [0.002]	− 0.042 (0.013) [0.001]	− 0.042 (0.013) [0.001]
Government_R&D (G)	0.001 (0.007) [0.889]	0.139 (0.019) [0.000]	− 0.004 (0.007) [0.553]	0.123 (0.022) [0.000]	0.189 (0.029) [0.000]	0.223 (0.032) [0.000]	− 0.009 (0.007) [0.241]	− 0.008 (0.007) [0.257]	− 0.008 (0.007) [0.256]
Innovativeness (I)			0.038 (0.019) [0.041]				0.032 (0.017) [0.060]	0.014 (0.030) [0.638]	0.035 (0.019) [0.066]
Suggestions (S)				0.068 (0.067) [0.312]	0.015 (0.050) [0.764]	0.098 (0.066) [0.139]	0.037 (0.022) [0.084]	0.107 (0.024) [0.000]	0.068 (0.041) [0.089]
Bonus_incentive (B)				0.038 (0.026) [0.146]	0.126 (0.033) [0.000]	0.131 (0.032) [0.000]	0.005 (0.010) [0.611]	0.001 (0.015) [0.932]	0.001 (0.015) [0.950]
G × S				− 0.036 (0.060) [0.551]		− 0.112 (0.083) [0.176]			
G × B					0.123 (0.062) [0.048]	0.141 (0.078) [0.071]			
I × S							0.179 (0.073) [0.015]		0.178 (0.074) [0.016]
I × B								0.014 (0.035) [0.696]	0.013 (0.034) [0.712]
Year effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	867	867	867	867	867	867	867	867	867
R ²	0.128	0.200	0.135	0.271	0.284	0.287	0.210	0.203	0.213

Note: Unstandardized regression coefficients are reported. Robust standard errors in parentheses. Two-tailed p-values in square brackets. The independent variable *government_R&D* is lagged by one year, resulting in $N = 3 * 289 = 867$ firm year observations.

firms, in turn, increase variation further through intra-firm processes that include knowledge development/sharing and employee participation, facilitating innovation. Simultaneously, selection mechanisms are at play, as not all firms deliver innovation, and within firms, not all innovations materialize into results. Finally, the two layers of variation are completed by the ultimate test of success or rejection by market forces, for example resulting in export success. This selective cycle serves to explain why, even in a sample of firms selected by their innovativeness, the overall export intensity may still be very low. The paths to the marketing of products of innovation are narrow, and export gateways are especially narrow for firms from a country with a large domestic market and considerable liabilities of origin due to possible poor country reputation. Indeed, Tan and Yang (2021) point out that Chinese firms experience significant challenges going abroad due to

their liability of origin. Thus, the analysis of variation-selection processes provides a unique perspective which we believe helps to reshape our understanding about why export success rates in emerging economies are low even for innovative firms.

Second, our study provides new empirical evidence for the R&D support-export relationship. We show that increased R&D activity leads to increased exports via innovation, through indirect-only mediation. This demonstrates that government support does not have to be targeted at specific desired outcomes to have positive socio-economic outcomes because one needs to consider potential spillover effects. Indeed, there are models that suggest that use stepwise logic to link R&D input innovation to increased productivity via innovation (Crepon et al., 1998; Halpern & Murakozy, 2012). In our exporting study, we demonstrate that government support for innovation results in increased innovation

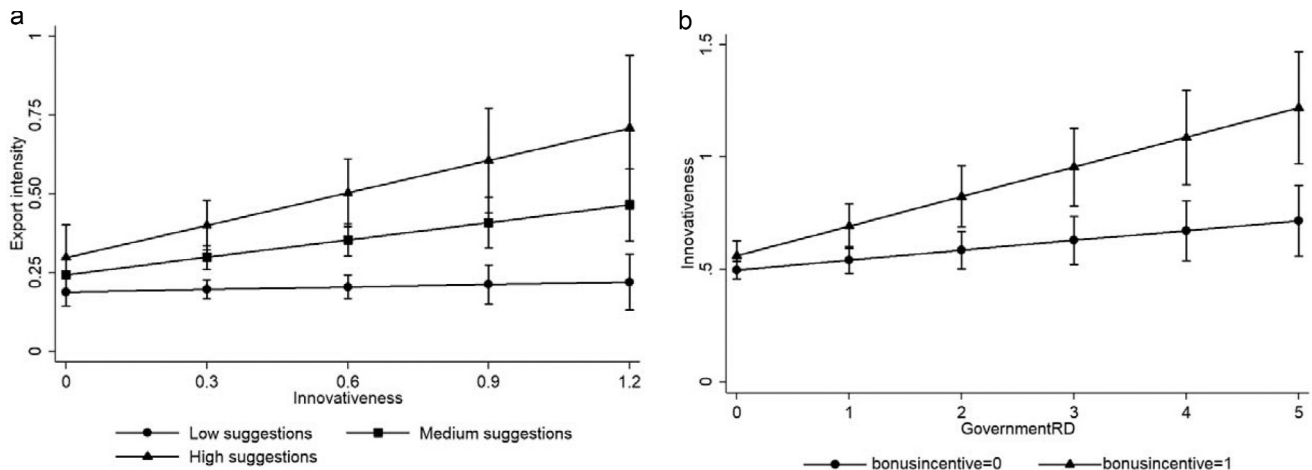


Fig. 3. a The moderating role of *suggestions* in the relationship between innovativeness and export intensity (H2b). b The moderating role of *bonus_incentive* in the relationship between government R&D support and innovativeness (H3a).

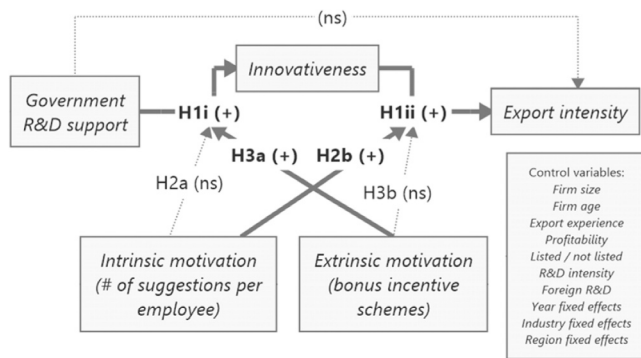


Fig. 4. Summary model showing results.

which in turn positively affects exports. Our results increase knowledge about the direct and indirect effects of government support policies highlighting the benefits of an integrated framework considering the combined effects of R&D influence on innovation and exports.

Third, by leveraging the KBV, our study enriches our understanding of boundary conditions by demonstrating that both intrinsic and extrinsic motivators moderate the mediated relationship between government R&D support and export performance. There has been much debate in the literature about whether intrinsic motivators, or extrinsic motivators better facilitate motivation and whether the presence of an extrinsic motivator could help or hurt the effectiveness of an intrinsic motivator (Gerhart & Fang, 2015). As Kuvaas et al. (2017, p. 245) state, “...there is an ongoing and somewhat politicized debate about whether these two types of motivation both have positive effects or whether they relate negatively and have differential effects.” While some research in psychology and economics suggests that financial incentives can harm intrinsic motivation (Frey & Oberholzer-Gee, 1997; Ryan & Deci, 2000a), experimental studies indicate this may not always be the case (e. g., Toubia, 2006). Similar to our research, there is growing evidence that both types of incentives can be helpful (Cerasoli, Nicklin, & Ford, 2014; Gerhart & Fang, 2015). For example, Cerasoli et al. (2014) meta-analysis found that both types of incentives have a positive effect on performance, with the positive effect of intrinsic incentives increasing when extrinsic incentives are present. Drawing on the work of Gagne and Deci (2005) and Self Determination Theory, Gerhart and Fang (2015) emphasized that providing employees with greater autonomy and not dictating how employees reach the outcome is a key factor in determining whether extrinsic incentives help or hinder motivation.

While incentives and employee motivation are integral to OB

Table 3

Robustness tests: exploring the effect of an instrumental variable.

Hypothesis	(1) H1 export-intensity	(2) H1 innovativeness	(3) H1 export-intensity
Dependent Variables			
Size	– 0.015 (0.005) [0.002]	– 0.091 (0.011) [0.000]	– 0.012 (0.005) [0.019]
Age	0.002 (0.012) [0.867]	– 0.040 (0.028) [0.154]	0.003 (0.012) [0.791]
Export_experience	– 0.001 (0.016) [0.955]	0.060 (0.036) [0.091]	– 0.003 (0.016) [0.863]
ROA	0.134 (0.036) [0.000]	0.109 (0.080) [0.173]	0.130 (0.036) [0.000]
Listed	0.032 (0.012) [0.008]	0.031 (0.027) [0.249]	0.031 (0.012) [0.010]
R&D_sales	0.312 (0.080) [0.000]	0.535 (0.179) [0.003]	0.296 (0.080) [0.000]
Foreign_R&D	– 0.044 (0.017) [0.010]	0.093 (0.039) [0.017]	– 0.047 (0.017) [0.006]
Government_R&D (G)	0.019 (0.012) [0.119]	0.177 (0.027) [0.000]	0.013 (0.013) [0.293]
Innovativeness (I)			0.031 (0.016) [0.050]
Year effects	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes
Region effects	Yes	Yes	Yes
N	867	867	867
Uncenter R ²	0.355	0.651	0.359

theories, they have not been saliently included in firm internationalization models. The findings of our study offer empirical evidence of extrinsic and intrinsic motivators as moderators paving the way for new and promising avenues in future firm internationalization research. The significance of contextual employee motivation effects in both stages of our model (government R&D support’s effect on innovation and innovation’s effect on exporting) indicates that it helps to have motivated employees both in transforming resources into innovation and subsequently into exports by developing, storing, and, if motivated, applying knowledge to exploit business opportunities. The benefits of employee suggestion systems also provide additional support for activation theory

Table 4
Regression analysis robustness tests of alternative measures.

Hypothesis	(1) H1	(2) H1	(3)	(5) H2a	(6) H3a	(7) Full model	(7) H2b	(8) H3b	(9) Full model
Dependent Variables	export-intensity	innovativeness	export-intensity	innovativeness	innovativeness	innovativeness	export-intensity	export-intensity	export-intensity
Size	− 0.009 (0.004) [0.028]	− 0.045 (0.009) [0.000]	− 0.008 (0.004) [0.062]	− 0.028 (0.010) [0.008]	− 0.028 (0.010) [0.008]	− 0.027 (0.010) [0.009]	− 0.013 (0.005) [0.003]	− 0.013 (0.005) [0.004]	− 0.013 (0.005) [0.004]
Age	0.003 (0.011) [0.751]	− 0.022 (0.028) [0.437]	0.004 (0.011) [0.694]	− 0.018 (0.026) [0.481]	− 0.016 (0.026) [0.535]	− 0.016 (0.026) [0.529]	0.008 (0.010) [0.433]	0.007 (0.010) [0.502]	0.008 (0.010) [0.439]
Export_experience	− 0.004 (0.018) [0.829]	0.027 (0.036) [0.456]	− 0.006 (0.018) [0.751]	0.025 (0.036) [0.487]	0.021 (0.036) [0.570]	0.021 (0.036) [0.563]	− 0.005 (0.017) [0.766]	− 0.007 (0.017) [0.664]	− 0.005 (0.017) [0.769]
ROA	0.133 (0.039) [0.001]	0.084 (0.077) [0.278]	0.130 (0.039) [0.001]	0.105 (0.073) [0.154]	0.109 (0.075) [0.145]	0.109 (0.075) [0.147]	0.147 (0.038) [0.000]	0.154 (0.039) [0.000]	0.147 (0.038) [0.000]
Listed	0.032 (0.013) [0.014]	0.038 (0.029) [0.188]	0.031 (0.013) [0.015]	0.009 (0.030) [0.757]	0.007 (0.030) [0.803]	0.007 (0.030) [0.817]	0.026 (0.013) [0.041]	0.025 (0.013) [0.048]	0.026 (0.013) [0.044]
R&D_sales	0.355 (0.105) [0.001]	0.855 (0.273) [0.002]	0.333 (0.104) [0.001]	0.880 (0.262) [0.001]	0.886 (0.263) [0.001]	0.887 (0.264) [0.001]	0.340 (0.100) [0.001]	0.338 (0.104) [0.001]	0.337 (0.101) [0.001]
Foreign_R&D	− 0.042 (0.013) [0.002]	0.143 (0.041) [0.001]	− 0.046 (0.013) [0.000]	0.110 (0.040) [0.006]	0.109 (0.040) [0.006]	0.109 (0.040) [0.006]	− 0.041 (0.013) [0.002]	− 0.042 (0.013) [0.001]	− 0.041 (0.013) [0.001]
Government_R&D (G)	0.028 (0.036) [0.437]	0.361 (0.083) [0.000]	0.016 (0.037) [0.669]	0.328 (0.108) [0.003]	0.376 (0.159) [0.018]	0.392 (0.177) [0.027]	0.014 (0.035) [0.695]	0.016 (0.035) [0.653]	0.014 (0.035) [0.684]
Innovativeness (I)			0.035 (0.018) [0.054]				0.038 (0.021) [0.064]	0.038 (0.022) [0.084]	0.036 (0.019) [0.058]
Suggestions (S)				0.069 (0.058) [0.238]	0.052 (0.051) [0.305]	0.066 (0.058) [0.253]	0.036 (0.031) [0.245]	0.106 (0.024) [0.000]	0.036 (0.031) [0.241]
Bonus_incentive (B)				0.037 (0.027) [0.165]	0.052 (0.031) [0.096]	0.051 (0.031) [0.099]	0.005 (0.010) [0.621]	0.003 (0.015) [0.870]	0.001 (0.015) [0.943]
G × S				− 0.113 (0.279) [0.685]		− 0.098 (0.272) [0.719]			
G × B					0.120 (0.073) [0.087]				
I × S							0.178 (0.073) [0.015]		0.178 (0.074) [0.016]
I × B								0.016 (0.035) [0.642]	0.015 (0.034) [0.657]
Year effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	867	867	867	867	867	867	867	867	867
R ²	0.129	0.172	0.135	0.255	0.256	0.257	0.209	0.202	0.212

and the advantages of activating employees (Scott, 1966) by providing more challenging tasks to enhance their effectiveness and prevent them from passively completing job duties.

6. Managerial and policy implications

How and under what conditions does government R&D support lead to innovation and in turn increased export performance? This question is most relevant for policymakers, but it is also important for managers of firms seeking to internationalize. Our study shows that government support for innovation helps firms to be more innovative which in turn helps firms to export more. Thus, one way to increase a firm's success in exporting is for a government to invest in R&D. Our study also shows that to achieve maximum innovation and export results it is important that employees are motivated and that firms consider both employee intrinsic and extrinsic motivation. For example, government R&D

support has a larger effect on a firm being innovative when the firm provides financial bonuses for successful innovations (extrinsic motivation). In contrast, for firms to achieve better export sales growth, it is most important to motivate employees to share their ideas about how to improve products and services (intrinsic motivation) and that systems exist to capture them.

Our study helps resolve this debate by helping managers understand which motivators more effectively boost innovation and which motivators boost exports. The moderating results we obtained are at first glance somewhat surprising. However, upon reflection more the results make sense. As shown in Fig. 3a, the extent of use of suggestion systems positively moderates the relationship between innovativeness and export intensity, but not the relationship between government support for R&D and innovativeness. This can be explained by the proposition that employee suggestions are often not directed at radical innovations (that would be patented) but towards incremental innovation and

process innovation that would be reflected more in improved marketing and financial performance.

In contrast, as seen in Fig. 3b, financial incentives moderate the relationship between government R&D support and innovativeness, but not the relationship between innovativeness and exporting. The finding, revealing that the relationship between government R&D support and innovation is not strengthened (positively moderated) by employees participating in suggestion systems but by using bonus financial incentives, is somewhat counterintuitive. This defies a view in the literature that innovation does not benefit from direct incentives, but from intrinsic motivation (e.g., suggestion systems). However, other studies like (Cerasoli et al., 2014) have like us found that extrinsic motivation can be beneficial. We suggest that bonuses are important for driving innovation since innovation requires many hours of hard work over a longer time period and some uncertainty of outcome exists, so a bonus can be helpful to give people a sustained extra push. Furthermore, to drive innovation bonuses are probably helpful in creating the feeling of being valued which motivates harder work. On the other hand, for exporting, there is perhaps a more direct and shorter-term correlation between effort invested and results obtained making a bonus less critical.

Our research suggests that the key to motivating employees depends on what one is trying to achieve. When aiming to encourage innovation, financial bonuses (extrinsic motivation) works best. The lack of impact from financial incentives on exporting can be attributed to the complexity of export sales, where incremental progress is often only visible to the sales personnel directly involved in export contracts and not to most employees. When aiming to boost export sales, it is crucial to motivate employees to share their ideas for improving products and services (intrinsic motivation) and have systems in place to capture these ideas. It is especially important to capture the insights of key employees who interact with overseas customers, as they have valuable knowledge about customer preferences that may be harder to understand due to cultural differences and limited interaction. The second part of our model focuses on channeling of innovative behavior towards facilitating exports and highlights the importance of a suggestion system as a key moderator in this process. Indeed, employees often have valuable ideas for product and service improvements that align with customer preferences but implementing these changes requires much effort and entails some risk. Therefore, it is important to ensure that employees are well-motivated for this to occur. Our findings have relevance for both the innovation and export literature, particularly in the current remote work environment where explicit incentives may play a larger role due to the distance between employees and their leaders.

For policymakers, helping firms to become more innovative and have more international sales are two key economic development goals in most countries, especially in emerging economies. Thus, it is appropriate that this study was set in China as these goals are especially important considering China's present stage of development with growing labor costs that require Chinese firms to add more value which innovation can help facilitate. We demonstrate that support for R&D activities, a discretionary financial resource, is transformed into core competencies and knowledge that delivers tangible export performance, i.e., it helps to improve innovation and there are positive spillovers to other activities such as exporting. This implies that government support mechanisms do not need to be targeted to specific areas and improvements may also appear in other performance metrics. Our study deepens our knowledge about government R&D support policies that need a unified framework to account for the joint effects on both innovation and exports, particularly in the context of emerging economies.

7. Limitations and future research

Like all studies our study has some limitations that also provide opportunities for future research. Similar to earlier researchers, we did

not establish a holistic model of export intensity. Instead, we focused on studying fragments of the theory. We incorporated aspects of time into traditionally cross-sectional research topics of government support and exporting. The processes related to innovation as its antecedents and the opportunities created by innovation take time. We have acknowledged this by linking the discussion to the evolutionary view and selecting variables that reflect its main processes of creating variation in combinations and narrowing the survival by selection.

Unfortunately, the data obtained from a secondary dataset did not allow us to develop more precise measurements for the constructs by including additional questions in the survey to either improve the framing of the question or add additional questions to synthesize latent variables. However, the level of detail in this dataset exceeded what is typically available in secondary datasets and still allowed us to delve deeper into the microfoundations of innovation and export relationships. We believe we selected the best available variables to align with our theoretical investigation. We encourage further studies to test our empirical models using purposefully collected multilevel datasets that combine innovation data with new product and service sales data.

Our study focuses on the most innovative firms in China, which may be considered a limitation as the sample does not represent more average firms and could thus result in selection bias. On the other hand, it can also be seen as a strength as these innovative firms are what China and most countries are striving to create. Understanding how to help them become even more innovative and successful in exporting are important issues to understand. Our analysis demonstrates that even this sample of the most innovative Chinese firms face challenges in exporting as they export only about 10 % of their sales on average. Our study adds value by being unique, as most studies on innovation and especially exports typically focus either on statistically average firms through random sampling or are population studies.

On the other hand, our study is limited to Chinese firms. We welcome future research that explores the mediating relationship between government R&D support and exports through innovation in 'average firms' or SMEs, as well as in the most innovative firms in other countries and ideally across multiple countries. This will help determine the extent to which firms in various settings could also benefit from a range of indirect spillovers similar to those we discovered. However, given the China's importance in the world economy and the extensive public discourse about the impact of Chinese government support on firms, particularly in R&D, the empirical focus of our study can be seen as highly relevant and a strength of the study.

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Data availability

Data will be made available on request.

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